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## Study of Friction and Wear of ABS/ZnO Polymer Composite Using Taguchi Technique

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### Abstract

The present study considers the tribological behavior of polymer composite material prepared using acrylonitrile-butadiene-styrene (ABS) as the base material and micron-sized zinc oxide (ZnO) as the filler. The experiment is carried out in dry condition on block-on-roller multi-tribotester (DUCOM) in room temperature based on Taguchi's  $L_{27}$  orthogonal array (OA). Filler content, normal load and sliding speed are considered as the design parameter and coefficient of friction and specific wear rate are considered as the responses. The optimal combination of parameter for minimum friction coefficient and wear rate is determined using Taguchi technique. The optimal parameter combination for minimum coefficient of friction (COF) is found as 5 wt% filler, 35 N load and 120 rpm speed and lowest specific wear rate is obtained at the filler content of 15 wt%, load of 35 N and speed of 120 rpm. Further, analysis of variance (ANOVA) is applied to investigate the influence of design parameter on the coefficient of friction (COF) and specific wear rate of the polymer composite. The results show that the friction coefficient and specific wear rate are significantly influenced by the increase of filler content, load and speed. The most influential factor which affects the tribological properties is normal load followed by filler content and speed. Scanning electron microscopy (SEM) observations are also carried out to identify the wear mechanisms for the worn surfaces at optimal parameter combination.

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## 1. Introduction

Over the past decades, polymer based composites are considered as one of the interest area that have attracted many researchers due to the development of new composite materials in engineering components with desired physical and mechanical properties (Chang et al., 2005). Polymers can be filled with organic fillers, inorganic fillers, and metallic particulate materials. Inorganic-filled polymer composites have become attractive in polymer field due to its various advantages such as easiness in processing, cost effectiveness and excellent performance over the metals as well as improved properties such as tensile modulus, strength, heat deflection temperature, hardness, fracture toughness etc. (Wang et al., 2008). Polymer composites has a special property of self lubrication and this made the composites suitable in tribological applications such as cams, seals, brakes, bearings etc (Rashmi et al., 2011).

Acrylonitrile – butadiene – styrene (ABS) is one of the engineering thermoplastic terpolymers widely used over the past decades and find applications in many fields like automotive, aerospace, business machines, computers, telephone handsets etc. Acrylonitrile gives chemical resistance and heat stability, butadiene gives toughness and impact strength and the styrene gives rigidity and easiness of processability, whereas neat ABS as well as other polymers has its limitation in tribology due to high friction coefficient and wear rate (Difallah et al., 2012). A lot of efforts have been made by researchers to improve the tribological properties of polymers by incorporating various fillers (Yu et al., 2000; Selvin et al., 2004; Cho and Bahadur, 2005; Zhang et al., 2009; Wang et al., 2009). Wang et al. (2012) have studied the mechanical and tribological properties of ABS filled with graphite and carbon black and found that the fillers can effectively decrease the COF and wear rate. Chang et al. (2007) have investigated polyether-ether ketone (PEEK) and polyether-imide (PEI) reinforced with short carbon fibres, sub-micro  $\text{TiO}_2$ , ZnS and graphite and reported that the conventional fillers enhance both the wear resistance and load carrying capacity of base polymers. Xiang and Gu (2006) have reported friction and wear behaviour of poly-tetra-fluoro-ethylene (PTFE) with ultra-fine kaolin particles. The incorporation of kaolin particles reduces the wear rate by two orders of magnitude as compared to the unfilled PTFE, but friction coefficient increases over unfilled PTFE at filler concentrations of 10 wt %. Further, Jiang et al. (2008) have revealed the tribological properties of polyphylene sulphide (PPS) reinforced with sub-micro  $\text{TiO}_2$  and short carbon fibres (SCF) and found that 15 vol % SCF and 6 vol %  $\text{TiO}_2$  gives the lowest coefficient of friction based on artificial neural network (ANN) prediction.

ZnO as a functional inorganic filler has great potential because of its prominent physical and chemical properties (Erjun et al., 2006). It has been widely used in areas such as optical materials, cosmetics and functional devices (Zhao et al., 2006; Tjong et al., 2006). Zinc oxide filled polymers are studied in many research articles related to the mechanical properties. ZnO particles can improve the mechanical properties of polymer composites (Lee et al., 2008). Mechanical properties of the HDPE/ZnO-Mg  $(\text{OH})_2$ - $\text{CaCO}_3$  polymer composites are investigated and it is found that tensile modulus and strength decrease with increasing filler content (Sezgin et al., 2012). It is seen that by adding 1 wt% of nano-ZnO filler into polypropylene (PP) matrix has enhanced the tensile strength, tensile modulus and elongation of the composites (Lin et al., 2009).

It is seen from the literature review, that there is a scarcity of literatures related to ABS filled with micron-sized ZnO filler in the field of tribology. In this experimental study, Acrylonitrile-Butadiene-Styrene (ABS) polymer material is selected as the matrix material and ZnO is added into ABS polymer as a filler material with different percentages of 5, 10 and 15 wt %. The tribological properties, coefficient of friction and specific wear rate are investigated for ABS/ZnO composites under different load conditions along with different sliding speed for a constant time of 300 sec at room temperature. Taguchi's  $L_{27}$  orthogonal array (OA) is used for conducting the tests. The experimental results are analyzed using Taguchi method and the optimal process parameter combination is determined for minimization of response values. A confirmation test was also carried out to verify the improvement of quality characteristics using the optimum design parameter with the initial parameter. The influence of design parameter and the interactions on COF and specific wear rate for the composites are also studied using analysis of variance (ANOVA). Finally, an effort was also made to study the morphology of wear tracks after tribological tests using scanning electron microscopy (SEM) images.

## 2. Experimental details

### 2.1. Materials

The matrix selected for this study is acrylonitrile-butadiene-styrene (ABS). It is Absolac-920 grade supplied in pellets form by Styrolution ABS limited, India with a density of  $1.04 \text{ g / cm}^3$  and melt flow index of 21 g/10 min. The filler selected is zinc oxide (ZnO) supplied by Central drug Ltd, India in the form of powder with a mean particle size of  $0.3 - 0.4 \text{ }\mu\text{m}$  and bulk density of  $5.61 \text{ g / cm}^3$ .

### 2.2. Sample preparation

ABS pellets and powders of ZnO are dried at  $60^\circ\text{C}$  in a vacuum oven for 6 hours to remove moisture. The materials are then weighed in the proportions needed and the mixture is extruded by using a Haake single screw extruder (Rheocord – 9000) with a screw diameter of 18mm and L/D ratio of 24:1. ABS / ZnO with different compositions are pre-mixed manually in a zip-lock bag before extrusion. The extruder is fitted with a rod die, screw speed of 60 rpm and die temperature of  $240^\circ\text{C}$  is employed for all the compositions (5, 10 and 15 wt% of ZnO filler). The temperature profile of the extruder is shown in Table 1 and the mixing for different compositions is carried out in a continuous manner. The extruded composites in the shape of rod are immediately cooled by water followed by air cooling. Then the composite rods are pelletized into granules form in uniform size by using a pelletizer machine. The pelletized composites are dried at  $60^\circ\text{C}$  in a vacuum oven for 6 hours to remove moisture before compression molding process. The pelletized granules are placed in a rectangular mold of size  $150 \times 100 \times 8 \text{ mm}^3$  and subjected to hot compression mold (Carver Press, Germany) with a temperature of  $260^\circ\text{C}$  and load of 8 metric tonnes kept for 1 min and then the load is lowered to 6 metric tonnes to allow the entrapped air out from the mold and kept for 15 min. Then, heat is turned off and the mold is allowed to cool in the compression machine itself at the room temperature for 2 hrs and kept at the same load up to removing the composite rectangular bar from the mold. The specimens for tribological tests are cut from the rectangular bar with a specimen size of  $20 \times 20 \times 8 \text{ mm}^3$ .

Table 1. Temperature profile along the extruder barrel.

| Feed Zone           | Compression Zone    | Metering Zone       | Die                 |
|---------------------|---------------------|---------------------|---------------------|
| $210^\circ\text{C}$ | $220^\circ\text{C}$ | $230^\circ\text{C}$ | $240^\circ\text{C}$ |

### 2.3. Design of experiments

The selection of design factors is the important stage for the design of experiments. There are many design factors such as filler content, load, speed, temperature, materials selected, sliding velocity etc., which will affect the test results of friction coefficient and wear rate. The key step in Taguchi method is to optimize the process parameter to achieve best quality performance. If the number of process parameter increases, there are lots of experiments have to be conducted to get the optimized parameter. To make the task easy, Taguchi method uses design of orthogonal arrays (OA) to study the process parameter with small number of experiments (Phadke., 1989; Bahadur and Tabor, 1985; Biswas and Satapathy, 2009; Rashmi et al., 2011; Siddharatha et al., 2011; Difallah et al., 2012). In the present study filler content, normal load, sliding speed is considered as the design parameter. The design parameter with levels is shown in Table 2. In order to study the effect of parameter and the interactions, a pre-designed orthogonal array,  $L_{27}$  is used in this study considering both the main factor effects and its interactions. The selection of orthogonal array is based on degrees of freedom for the experiments (Taguchi and Konishi, 1987). The main factor has 2 degrees of freedom and for two way interaction of the factors, the degrees of freedom is 4. Therefore, total degrees of freedom will be  $(3 \times 2) + (3 \times 4) = 18$ . The total degrees of freedom of the OA should be greater than the experimental DOF of the factors according to Taguchi method. So,  $L_{27}$  (26 DOF), is chosen for this study.

Table 3 shows the details of design parameter. The complete table for L<sub>27</sub> OA is omitted here for brevity. The tests are conducted as per the experimental design given in Table 3 at room temperature.

#### 2.4. Friction and wear tests

Tribological tests for coefficient of friction (COF) and wear rate of ABS / ZnO composite are performed on a multi-tribotester block TR25 (Ducom, India) under dry condition at room temperature. The composite samples (20 X 20 X 8 mm<sup>3</sup>) are pressed against a rotating steel roller (diameter 50 mm, thickness 50 mm and material EN8 steel) of hardness 55 HRC. The rotating steel roller serves as a counter face and the stationery block serves as the test specimen. The surfaces of the specimen and roller are cleaned with a soft paper before each test to ensure proper contact with the counter face. A loading lever is used to apply a normal load on the top of the specimen. The frictional force is measured by a force sensor. Tests are carried out based on L<sub>27</sub> OA and each run is conducted for 300 sec. The experimental data of coefficient of friction is recorded on a computer attached to the testing apparatus. The weight loss is used to calculate the specific wear rate. The samples were weighed before and after the experiments to an accuracy of 0.0001 g in a mettler toddler electronic balance. The specific wear rate (W<sub>s</sub>) is calculated using equation (1) (Wang et al., 2010).

$$W_s = \frac{W_1 - W_2}{\rho * P * v * t} \quad (1)$$

where W<sub>s</sub> is the specific wear rate in mm<sup>3</sup> / N.m, W<sub>1</sub> is the weight before the test in g, W<sub>2</sub> is the weight after the test in g, ρ is the computed density of composites in g / cm<sup>3</sup>, P is the applied normal load in N, v is the relative sliding velocity in m / s and t is the experimental time in sec.

Table 2. Design factors with different levels.

| Design factors  | Unit | Levels |      |     |
|-----------------|------|--------|------|-----|
|                 |      | 1      | 2    | 3   |
| % of filler (A) | %    | 5      | 10*  | 15  |
| Load (B)        | N    | 15     | 25*  | 35  |
| Speed (C)       | rpm  | 80     | 100* | 120 |

\* initial test conditions

#### 2.5. Taguchi method

Taguchi method is one of the powerful statistical tools used in the application of design and analysis for experiments adopted to optimize the design parameter and it is an effective approach to produce high-quality products at a relatively low cost (Taguchi., 1990; Ross., 1995). Taguchi has used loss function to measure the performance characteristic deviating from the desired value. The value of loss function is then transformed into signal-to-noise (S/N) ratio to calculate the quality characteristics and evaluate numerous parameter. The S/N ratio is the ratio of the mean (signal) to the standard deviation (noise). Usually, the standard S/N ratio has three categories of the performance characteristics, namely, smaller-the-better, nominal-the better and higher-the-better. The S/N ratio of each level of design parameter is computed based on the S/N analysis. In the present study, S/N ratio is calculated as the logarithmic transformation of the loss function by using smaller-the-better criterion (equation 2) as minimum values of friction coefficient and wear rate are required (Roy, 1990).

$$\frac{S}{N} \text{ ratio} = -10 * \log_{10} \left( \frac{1}{n} * y^2 \right) \quad (2)$$

where y represents experimental data for COF and specific wear rate and n denotes the number of experiments.

## 2.6. Analysis of Variance (ANOVA)

ANOVA is a statistical technique used to predict the process parameter and their interactions significantly affect the quality characteristics (Liao et al., 2004). This is done by separating the total variability of S/N ratio, which is measured by the sum of squared deviations from the total mean of S/N ratio, into contributions for each process parameter and the error. The percentage contribution can be used to determine the significant parameter which affects the performance characteristics. Also, F-test named after Fisher (1925) can be used to determine the parameter which has a significant effect on the quality characteristics based on 95% confidence level. Usually, when the F value is large the parameter has a significant effect on the performance. Using Minitab 16 software (2001), ANOVA is performed to determine the design parameter and its interaction significantly affect the performance characteristics.

## 3. Results and Discussions

### 3.1. Taguchi analysis

The tribological experiments for friction coefficient and wear rate are conducted as per the orthogonal array based design of experiments given in Table 3 and the experimental data for coefficient of friction and specific wear rate are shown in Table 3. Taguchi analysis is carried out using Minitab 16 statistical software. S/N ratio is calculated based on smaller-the-better criterion for COF and specific wear rate. S/N ratios for both the responses are shown in Table 4. Influences of each design parameter (A, B and C) on coefficient of friction and specific wear rate are obtained from the response tables of mean S/N ratio. Response tables for COF and specific wear rate are presented in Table 5 and Table 6 respectively. The main effect plots for mean S/N ratios of COF and specific wear rate are presented in Fig. 1. In the main effects plots, if the point is near the average horizontal line has less significant effect and the one which has highest inclination will have most significant effect on the responses. Taguchi recommends that the larger S/N ratio corresponds to the best quality characteristics, regardless of the category of the performance characteristic. Therefore, the optimal level of process parameter is the level with the highest S/N ratio. From response table for COF (Table 4) and main effects plot (Fig. 1), the optimum combination of design parameter is A1B3C3. Similar results have been reported in previous research which analyzed the friction and wear loss of ultra high molecular weight polyethylene filled with micro and nano-ZnO (Chang et al., 2013). For specific wear rate, the optimum combination is A3B3C3 and the results are in agreement with previous research (Cho et al., 2005; Rashmi et al., 2011; Wang et al., 2012).

From this study, it is also clear that COF decreases when the filler loading is at 5 – 10 wt% for high loads and speed because the surface temperature of the composites increases with the high load and the composite surfaces become soft caused by frictional heat at the interface, thus the reduction in COF occurs (Chang et al., 2013). Hence, as the applied load increases, the frictional coefficient reduces. In the case of high speed, due to shear action between the roller and specimen, thin layer of polymer debris will be formed at the interface and the frictional contact takes place at the accumulated debris. After a certain number of rotations, the friction will reach the steady state due to polymer debris at the interface. Hence, the COF reduces with an increase in rotational speed (Chang et al., 2013).

In case of specific wear rate, with the addition of filler it decreases for the speed varied and load applied. This may be attributed to the detachment of surfaces from the bulk material which is caused by the shear stress formed at the rotational action of wear. The particle filled polymer matrix withstands the shear action due to increase in modulus and it cannot be easily detached from the composite (Felhos et al., 2008). Hence, increase in wear resistance is noted after the incorporation of filler with ABS polymer. The minimum wear rate is found at 15 wt% filler, 35 N load and 120 rpm. The results are further supported by the investigation of tribological properties of polymer reported by Yu and Bahadur, 1998. At high speed and load, the wear rate is decreased due to softness formed in composite surface caused by frictional heat (Zhao and Bahadur, 2002).

### 3.2. Analysis of variance (ANOVA)

In order to understand the effect of design factors like % of filler (A), load (B) and speed (C) and the interactions on the experimental data, analysis of variance (ANOVA) is studied at 95% level of confidence. The results of ANOVA for COF and specific wear rate are presented in Table 6 and Table 7 respectively. The table shows the F – ratio and percentage contribution of each factor which affects the performance characteristics. It is seen that normal load is the most significant factor followed by filler content and other parameter and interaction terms both for COF and specific wear rate. Contributions of normal load on COF and specific wear rate are about 77% and 71% respectively. The results of ANOVA are supported by previous studies of Cho et al. (2005), Rashmi et al. (2011) and Siddhartha et al. (2011).

Table 3. Experimental data and S/N ratio for COF and specific wear rate

| Expt. Run | 1-% of filler (A) | 2-Load (B) | 3-Speed (C) | Coefficient of friction | Specific wear rate (mm <sup>3</sup> /Nm) | S/N ratio (COF) | S/N ratio (sp. wear rate) |
|-----------|-------------------|------------|-------------|-------------------------|------------------------------------------|-----------------|---------------------------|
| 1         | 1                 | 1          | 1           | 0.3944                  | 0.002682                                 | 08.08126        | 51.42999                  |
| 2         | 1                 | 1          | 2           | 0.3508                  | 0.002685                                 | 09.09881        | 51.42167                  |
| 3         | 1                 | 1          | 3           | 0.3466                  | 0.001953                                 | 09.20343        | 54.18729                  |
| 4         | 1                 | 2          | 1           | 0.2695                  | 0.001961                                 | 11.38882        | 54.14867                  |
| 5         | 1                 | 2          | 2           | 0.2483                  | 0.003141                                 | 12.10047        | 50.05795                  |
| 6         | 1                 | 2          | 3           | 0.2152                  | 0.001205                                 | 13.34315        | 58.37958                  |
| 7         | 1                 | 3          | 1           | 0.2056                  | 0.001365                                 | 13.73809        | 57.29685                  |
| 8         | 1                 | 3          | 2           | 0.1976                  | 0.001410                                 | 14.08617        | 57.01848                  |
| 9         | 1                 | 3          | 3           | 0.2097                  | 0.000909                                 | 13.56871        | 60.83252                  |
| 10        | 2                 | 1          | 1           | 0.4379                  | 0.002912                                 | 07.17191        | 50.71593                  |
| 11        | 2                 | 1          | 2           | 0.4017                  | 0.002048                                 | 07.92289        | 53.77545                  |
| 12        | 2                 | 1          | 3           | 0.4787                  | 0.002742                                 | 06.39906        | 51.23872                  |
| 13        | 2                 | 2          | 1           | 0.2267                  | 0.001577                                 | 12.88963        | 56.04455                  |
| 14        | 2                 | 2          | 2           | 0.2804                  | 0.002048                                 | 11.04510        | 53.77545                  |
| 15        | 2                 | 2          | 3           | 0.2102                  | 0.001163                                 | 13.54607        | 58.68858                  |
| 16        | 2                 | 3          | 1           | 0.2075                  | 0.001096                                 | 13.65861        | 59.20510                  |
| 17        | 2                 | 3          | 2           | 0.2230                  | 0.001048                                 | 13.03351        | 59.59166                  |
| 18        | 2                 | 3          | 3           | 0.1600                  | 0.001033                                 | 15.91861        | 59.71541                  |
| 19        | 3                 | 1          | 1           | 0.5125                  | 0.002403                                 | 05.80666        | 52.38418                  |
| 20        | 3                 | 1          | 2           | 0.4013                  | 0.002122                                 | 07.93101        | 53.46599                  |
| 21        | 3                 | 1          | 3           | 0.4848                  | 0.002338                                 | 06.28891        | 52.62365                  |
| 22        | 3                 | 2          | 1           | 0.2863                  | 0.001294                                 | 10.86394        | 57.76109                  |
| 23        | 3                 | 2          | 2           | 0.3766                  | 0.001747                                 | 08.48149        | 55.15529                  |
| 24        | 3                 | 2          | 3           | 0.2723                  | 0.001378                                 | 11.29948        | 57.21436                  |
| 25        | 3                 | 3          | 1           | 0.2683                  | 0.000898                                 | 11.42665        | 60.93543                  |
| 26        | 3                 | 3          | 2           | 0.2271                  | 0.000994                                 | 12.87708        | 60.05294                  |
| 27        | 3                 | 3          | 3           | 0.2747                  | 0.000721                                 | 11.22290        | 62.84500                  |

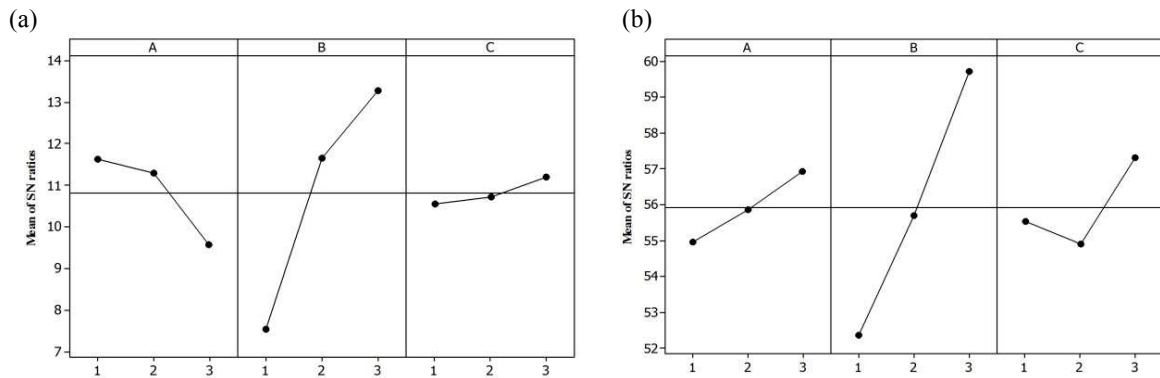


Fig. 1. Main effects plot for S/N ratio (a) COF (b) Specific wear rate

### 3.3. Confirmation test

From Taguchi analysis, the optimal levels of design parameter are selected for COF and specific wear rate. The final step is to predict and verify the improvement of the response using the optimal level of process parameter. Confirmation tests are carried out to validate the experimental results and to evaluate the accuracy of the analyses. Using the optimal level of testing parameter, the estimated S/N ratio,  $\bar{\gamma}$  is calculated using equation (3) (Roy, 1990).

$$\bar{\gamma} = \gamma_m + \sum_{i=1}^0 (\gamma_i - \gamma_m) \quad (3)$$

where  $\gamma_m$  is the total mean S/N ratio,  $\bar{\gamma}_i$  is the mean S/N ratio at the optimal testing parameter level and 0 is the number of main design process parameter that significantly affect the performance of polymer composites.

Table 8 and Table 9 show the results of confirmation tests for COF and specific wear rate respectively. The improvements of S/N ratios from initial to optimal levels are about 19% and 14% for COF and specific wear rate respectively. Therefore, friction and wear performances are improved by using Taguchi method.

Table 4. Response table for each factor levels for COF

| Average S/N ratio for each factor level(COF) |         |         |         |
|----------------------------------------------|---------|---------|---------|
| Level                                        | A       | B       | C       |
| 1                                            | 11.623* | 7.545   | 10.558  |
| 2                                            | 11.287  | 11.662  | 10.731  |
| 3                                            | 9.578   | 13.281* | 11.199* |
| Delta                                        | 2.046   | 5.736   | 0.641   |
| Rank                                         | 2       | 1       | 3       |

“\*” indicates optimal process level

Table 5. Response table for for specific wear rate

| Average S/N ratio for each factor level (Specific wear rate) |        |        |       |
|--------------------------------------------------------------|--------|--------|-------|
| Level                                                        | A      | B      | C     |
| 1                                                            | 54.97  | 52.36  | 55.55 |
| 2                                                            | 55.86  | 55.69  | 54.92 |
| 3                                                            | 56.94* | 59.72* | 57.3* |
| Delta                                                        | 1.96   | 7.36   | 2.38  |
| Rank                                                         | 3      | 1      | 2     |

“\*” indicates optimal process level

Table 6. ANOVA table for COF

| Source of variation | Degrees of freedom | Sum of squares | Mean squares | F - ratio | % contribution |
|---------------------|--------------------|----------------|--------------|-----------|----------------|
| A                   | 2                  | 21.662         | 10.831       | 11.03*    | 10.58852       |
| B                   | 2                  | 157.431        | 78.716       | 80.15*    | 76.95327       |
| C                   | 2                  | 1.978          | 0.989        | 1.01      | 0.96686        |
| AxB                 | 4                  | 4.781          | 1.195        | 1.22      | 2.33698        |
| AxC                 | 4                  | 2.254          | 0.563        | 0.57      | 1.10177        |
| BxC                 | 4                  | 8.618          | 2.154        | 2.19      | 4.21253        |
| Error               | 8                  | 7.857          | 0.982        |           | 3.84055        |
| Total               | 26                 | 204.5800       |              |           | 100.0000       |

Significant at 95% confidence level ( $F_{0.05,2,8} = 4.46$  &  $F_{0.05,4,8} = 3.84$ )

“\*” indicates most significant factor.

Table 7. ANOVA table for specific wear rate

| Source of variation | Degrees of freedom | Sum of squares | Mean squares | F - ratio | % contribution |
|---------------------|--------------------|----------------|--------------|-----------|----------------|
| A                   | 2                  | 17.39000       | 8.695        | 10.98*    | 5.05570        |
| B                   | 2                  | 244.57200      | 122.286      | 154.45*   | 71.10313       |
| C                   | 2                  | 27.39200       | 13.696       | 17.3*     | 7.96353        |
| AxB                 | 4                  | 7.16500        | 1.791        | 2.26      | 2.08304        |
| AxC                 | 4                  | 16.78000       | 4.195        | 5.3*      | 4.87836        |
| BxC                 | 4                  | 24.33400       | 6.084        | 7.68*     | 7.07450        |
| Error               | 8                  | 6.33400        | 0.792        |           | 1.84145        |
| Total               | 26                 | 343.96800      |              |           | 100.0000       |

Significant at 95% confidence level ( $F_{0.05,2,8} = 4.46$  &  $F_{0.05,4,8} = 3.84$ )

“\*” indicates most significant factor.



### 3.4. Scanning electron microscopy (SEM) analysis

SEM examinations are carried out for the composite surfaces coated with thin platinum film on the worn out surface by sputtering to get a conducting layer on a JEOL (model JSM 6390LV, Japan) microscope to observe the morphology of wear tracks of ZnO filled with the ABS polymer matrix after the tribology tests. Fig. 2 (a) shows the SEM images for initial condition. From the micrographs, it is observed that worn surfaces are mainly composed of longitudinal groove that is caused by micro-cutting and micro-ploughing sliding action. From the Fig. 2(b-c) for the worn surface of ZnO filled ABS composites, fillers covered the matrix region which results in reduced COF and wear rate. Fig 2 (b) shows the micrographs for the optimal condition of friction coefficient and it is revealed that the polymer debris is formed on the surface, due to this the frictional contact between composite and roller will take place at the debris. Hence, with the increase in load and speed, friction coefficient decreases. Fig. 2 (c) shows the image for optimal condition of wear rate. It reveals that the composite surface is smooth and small amount of debris is formed and due to the generation of frictional heat, the composite surface becomes smooth with high load and speed which supports the wear rate data shown in Table 3.

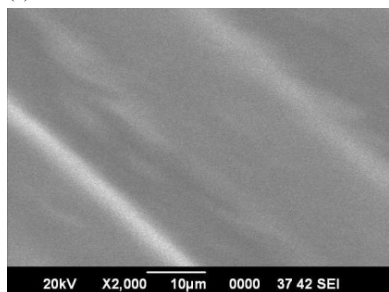
Table 8. Confirmation test for estimated and actual S/N ratio of COF

|                | Initial Parameter | Predicated Optimal | Expt.,  | Improvement in result |
|----------------|-------------------|--------------------|---------|-----------------------|
| Level          | A2B2C2            | A1B3C3             | A1B3C3  |                       |
| COF            | 0.2804            |                    | 0.2097  |                       |
| S/N ratio (dB) | 11.0451           | 14.0747            | 13.5687 | 18.60%                |

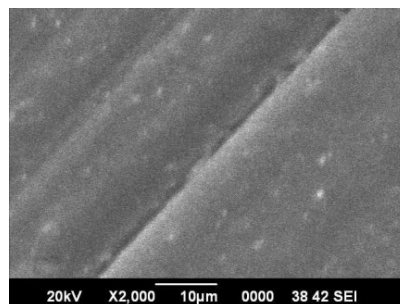
Table 9. Confirmation test for estimated and actual S/N ratio of specific wear rate

|                | Wedge parameter | Predicated Optimal | Expt.,  | Improvement in result |
|----------------|-----------------|--------------------|---------|-----------------------|
| Level          | A2B2C2          | A3B3C3             | A3B3C3  |                       |
| COF            | 0.00205         |                    | 0.00072 |                       |
| S/N ratio (dB) | 53.7755         | 62.1133            | 62.8450 | 14.43%                |

(a)



(b)



(c)

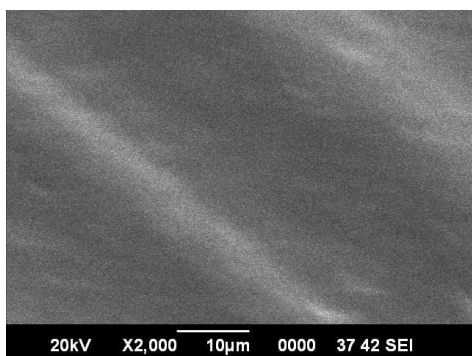


Fig. 2. SEM images: (a) after the test at initial condition (A2B2C2) (b) after the test at optimal condition for COF (A1B3C3) (c) after the test at optimal condition for wear rate (A3B3C3)



#### 4. Conclusions

The tribological property of friction coefficient and specific wear rate has been carried out for ABS matrix filled with ZnO filler using Taguchi analysis. It can be concluded that with the addition of filler the friction coefficient and wear rate decrease with an increase in load and speed. The optimal condition for coefficient of friction of polymer composites is found to be 5 wt% filler content, 35 N applied load and 120 rpm speed. In the case of specific wear rate, it is found to be 15 wt% filler content with the applied load of 35 N and speed of 120 rpm. SEM images are used to support the results. It can also be concluded that the design factor, applied load has the major contribution on tribological property followed by filler content and speed. The confirmation test show the improvement of COF (S/N ratio) from initial to optimal condition by 18.60% and for wear rate (S/N ratio) by 14.43%. It can be concluded from this study that with the addition of micron-sized ZnO filler with the ABS matrix at the right combination of load and speed, the tribological properties get improved.

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